



Optimization Analysis Controlling Motorcycle Spare Part Inventory Using the Always Better Method Control and Economic Order Quantity (Case Study: AHASS Arie Indah Motor)

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ABSTRACT

AHASS Arie Indah Motor often faces problems in inventory management. This inventory problem is caused by several factors, such as unstable product inventory and long waiting times. The implementation used is the ABC and EOQ methods. The research method used is the quantitative descriptive method.

With the ABC method, spare parts in category A have an investment value percentage of around 80%, category B around 15% and category C around 5%. With the number of spare part items in category A 54 types with a quantity of 11,213 items, category B 92 types with a quantity of 3,490 items and category C 258 types with a quantity of 830 items. With the EOQ method, the results obtained at AHASS Arie Indah Motor for the type of spare part unit (OLI MPX 10W30 SL 0.8L IDE) are optimal purchases of 122 units, ordering frequency of 15 units, total safety stock of 59 units, number of reorder points of 112 units and maximum inventory of 182 units.

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1. INTRODUCTION

Inventory management is one part of management control. Broadly speaking, inventory refers to existing stock and provided to meet consumer demand (Meirizha, 2022). Inventory control can be said to be effective if it can be fulfilled customer requests immediately and optimize costs (Supriyadi & Nurdewanti, 2022).

AHASS Arie Indah Motor is an official automotive company specializes in providing service and provider care solutions motorbike spare parts. In AHASS framework, spare part availability is categorized into two segments The main ones are Honda Genuine Parts (HGP) and Non HGP, which are generally known as authentic spare parts supported by PT. Astra Honda Motor (AHM).

Problems or obstacles that are often encountered at AHASS Arie Indah Motors are the fulfillment of demand for spare parts supplies, namely unstable supply of goods and quite long lead times. Instability of demand for spare parts, as well as unexpected needs too will cause excess or shortage of inventory resulting in costs (Hidayat, 2020). When there is a shortage of inventory, it is necessary AHASS makes purchases to other agents to fulfill demand. This problem requires proper control because of the waiting time (lead time) and reorder point (reorder point) are very influential regarding the amount of costs incurred (Sofiana & Ahmad Tasdiqul Haq, 2020).

There are methods that can be used to determine levels inventory, such as Always Better Control (ABC) and Economic Order Quantity (EOQ). ABC analysis is a method for allocating costs or level of control and frequency of review of goods (Al-Fikri & Ghofari, 2023). Application of classification ABC is widely used to categorize goods based on quantity, where inventory is usually separated into three categories. Group A represents The highest investment value is around 80%, followed by group B with value investment of around 15% and group C with an investment value of around 5% (Setiani, 2021).

Economic Order Quantity (EOQ) provides a direct approach to determining order quantity optimality, safety stock, and reorder point. This technique involves calculation of order quantities for each order to minimize costs overall inventory (Ramadhan, 2018). Factors that considered in the EOQ calculation usually includes ordering costs, storage costs, and purchasing costs, with this method also facilitate determination of stock levels required to meet demand (Umami, 2018).

It can be concluded that the ABC classification focuses on grouping components that have high or low values and their movement cycles so that inventory control is more effective and efficient. Whereas The use of the EOQ method in inventory control is used for determine the optimal point of ordering and minimize costs. These two methods interrelated where ABC is used for grouping and EOQ for cost

optimization materials, semi-finished materials, and finished goods, are important to meet customer needs and satisfaction, as well as anticipate urgent needs.

2. LITERATURE REVIEW

2.1 Inventory Management

Inventory management is an important thing and must be paid attention to in an organization or company. Inventory management plays a role control center for raw materials or parts which involve activities reception, storage, maintenance and distribution (Lutfiana & Puspitosari, 2020).

Heizer and Render (2015) explain that there are several functions of inventory as a company's operating flexibility, here are some function:

1. Provide options to meet customer requests and also avoids fluctuations.
2. As a separator of several process stages.
3. To benefit from discounts due to purchasing in bulk big.
4. To avoid inflation and price increases.

Hakim (2019), mentions several benefits of inventory as a basis meeting company needs, as follows:

1. Minimize the risk of delays in delivery of goods.
2. Minimize materials or goods ordered that are not in good condition which requires it to be returned.
3. Anticipate an increase in goods prices or inflation.
4. Store goods in anticipation of seasonal goods so that it doesn't make things difficult for the company when the goods come to market emptiness.
5. Get profits and quantity discounts based on purchases.
6. The availability of necessary goods will provide service satisfaction towards customers

2.2 Always Better Control (ABC)

ABC analysis involves categorization of items according to the level of capital absorption or share allocation based on the amount of investment tied up in the inventory of each part (Al-Fikri & Ghofari, 2023).

The classification of goods is divided into 3 categories, as follows (Guslan & Saputra, 2020)

1. Category A (80-20) means the absorption of funds is around 80% of the total capital or investment value of goods and 20% is the amount of goods from all types of goods managed.
2. Category B (15-30) absorbs funds around 15% of all capital or investment value and 30% is the type of goods from all existing goods.
3. Category C (5-50) is the remainder of categories A and B, namely only absorbing funds around 5% and the number of types of goods is 50% of the total

2.3 Order Cost

Ordering costs are costs arising from ordering activities goods, these costs are variable with order frequency. It means more and more. The larger or higher the order level, the greater the costs the order. Ordering costs usually include shipping costs, order processing costs, and costs of placing goods in the warehouse (Mayasari, 2021).

Ordering costs by the procurement department are calculated based on The total costs incurred to order goods in a year are divided with PO within one year so that the value of the ordering costs is obtained every time message. Ordering costs can be determined using the following formula:

$$\text{Order Cost} = \frac{\text{total annual order value}}{\text{total items of goods}}$$

2.4 Holding Cost

Storage costs or holding costs are costs incurred on inventory storage. These costs are incurred in an effort to protect the goods from damage, obsolescence, and risk of loss of goods. The following are: costs related to storage costs (Tarigan, 2023):

- a. Warehouse facilities (electricity, lighting, cooling, and other equipment).
- b. Cost of obsolescence of goods.
- c. Goods handling costs.
- d. Cost of loss.
- e. Rental or capital costs.
- f. Insurance costs.

The calculation of storage costs is determined in the percentage value of the costs incurred for the warehouse from the total inventory that has been set. Then the percentage of holding cost of storage value is multiplied by the quantity per group divided by the total storage to obtain the storage cost.

$$\text{Storage Value} = \frac{\text{total storage}}{\text{qty of each group}}$$

$$\text{Percentage} = \frac{(\text{storage value} \times \text{qty of each group})}{\text{total storage}}$$

2.5 Economic Order Quantity (EOQ)

Economic Order Quantity is a strategic approach that used in inventory management to improve quantity efficiency orders and reduce costs (Safitri, 2021). According to Vikaliana (2020), EOQ is the optimization of purchases or quantities obtained at the most economical cost possible. This calculation focuses on achieving optimal quantity by minimizing inventory costs (Al-Fikri & Ghofari, 2023). The formula for determining optimum ordering is as follows:

$$\text{EOQ} = \sqrt{(2SD/H)} / \sqrt{(2SD/I \cdot C)}$$

Description:

- Q = Economical quantity per order
 D = The number needs in units (unit) per year
 S = Ordering Cost for Once Order
 H = Storage cost per unit per year
 I = % storage cost
 C = Purchase price

2.6 Safety Stock

Haizer and Rander (2015) define safety stock as additional inventory or reserves held to anticipate varying demand. The following formula can be used to calculate safety stock:

$$\text{SS} = \frac{\text{LT}}{\text{Work Days}} \times \text{D}$$

Description :

- SS: Safety Stock
 LT: Lead Time (waiting time)
 Working Days: 26 Days (1 month)
 D : Quantity required

2.7 Reorder Point (ROP)

Reorder Point (ROP) is the point at which a company must reorder in order to maintain the availability of goods (Al-Fikri & Ghofari, 2023). If this activity is not carried out, it will have an impact on customer demand not being met due to stock shortages and also loss of profit for the company. The reorder point can be influenced by several factors as follows (Safitri, 2021):

1. Lead Time
2. Safety Stock

$$ROP = (LT \times D) + SS$$

Description:

SS = Safety Stock

LT = Lead Time

D = Number of Requests

2.8 Maximum Inventory

Maximum Inventory is needed by companies so that the amount of inventory in the warehouse is not excessive so that there is no waste of working capital. The formula used to calculate Maximum Inventory is as follows:

$$Max = EOQ + SS$$

EOQ = Optimal purchase quantity

SS = Safety stock

3. RESEARCH METHOD

The object of the study was conducted at the AHASS Arie Indah Motor company, which is an automotive company engaged in the field of service and provision of motorcycle spare parts located precisely at Jl. Raya Kudus - Jepara No.Km.5, Madaran, Mijen, Kec. Kaliwungu, Kab. Kudus, Central Java.

The method used is a quantitative descriptive method. Descriptive research is research conducted to analyze and present data systematically or describe existing data so that it is easier to understand and conclude (Sudaryana, 2022). The purpose of using this type of quantitative descriptive research is to present data systematically, factually, and accurately (Sugiono, 2019). While quantitative is research that involves the use of numbers or figures processed with statistics (Abdullah, 2015). The main purpose of using quantitative is to find out the value or numbers in fact from the processing of existing data.

4. RESULT AND DISCUSSION

4.1 Grouping Based on ABC Classification

The steps that need to be taken in carrying out ABC analysis are as follows:

- a. Calculate the number of spare parts needed per month or per year for each unit.
- b. Find the price of each unit and then make a list.
- c. Multiply the unit requirements by the unit price to get the investment value.
- d. Sort the investment value from the largest to the smallest, then make a percentage of the investment value.
- e. Calculate the cumulative investment value and group it.

The following are the results of the identification of spare part needs data at AHASS Arie Indah Motor during 2023 :

Table 1. ABC Classification

Grouping based on ABC				
Group	Amount	Quantity	Investment Value	%
A	53	11.213	735.350.775	80
B	92	3.490	139.085.790	15
C	258	830	46.521.100	5
Total	403	15.533	920.956.665	100

From the table above, the results of the ABC analysis are in accordance with the ABC percentage determination. It can be seen the difference in spare parts from each group. Where spare parts with category A have a percentage of investment value ranging from 80% with a number of items of 53, category B is around 15% with a number of items of 92 and category C is around 5% with 258 types of items. With the number of spare part items in category A 11,213 items, category B 3,490 items and category C 830 items. From the results of the table analysis, it shows that spare parts with category A have the highest value level, so that these spare parts are a priority for ordering so that there is no shortage of spare parts when needed.

4.2 Order Cost

The ordering cost is known in advance, which is IDR 12,000,000 within one year, which includes shipping, telephone and courier costs, then divided by the total of all types of goods ordered.

The following are the results of the identification of ordering costs at AHASS Arie Indah Motor during the 2023 period.

Calculation :

$$\text{Order Cost} = \frac{\text{total order value per year}}{\text{total item of goods(A, B, C)}}$$

$$\text{Order Cost} = \frac{12.000.000}{403}$$

$$\text{Order Cost} = \text{IDR } 29.777$$

Table 2. Identify Order Costs

Ordering Cost						
No	Criteria	Category	Qty	Monthly Value (IDR)	Annual Value (IDR)	Ordering Cost (IDR)
1	Shipping	A	53	400.000	4.800.000	29.777
2	Call	B	92	300.000	3.600.000	
3	Courier	C	258	300.000	3.600.000	
Total			403	1.000.000	12.000.000	

4.3 Storage Cost

Storage costs are calculated as a percentage of costs incurred for the warehouse from the total inventory value which has been determined with a storage value of 93,201. This value is obtained from dividing the total annual storage costs divided by the total of all types of goods. The following are the results of the calculation identification of storage costs:

$$\text{Storage value} = \frac{\text{Total Storage Cost}}{\text{Total Overall Qty}}$$

$$\text{Storage value} = \frac{37.560.000}{403}$$

$$\text{Storage value} = \text{IDR.93,201}$$

Then, to determine the percentage value, the following can be obtained:

$$\text{Percentage} = \frac{(\text{Storage Value} \times \text{Qty Each Group})}{\text{Total Storage Per Year}}$$

$$\text{Percentage} = \frac{(93.201 \times 53)}{37.560.000}$$

$$\text{Percentage} = 13 \%$$

This storage cost calculation focuses on category or group A, which are prioritized items.

Table 3. Storage Cost

Storage Cost						
No	Item	Annual Value (IDR)	Category	Qty	Storage Value (IDR)	%
1	Shrinkage	6.000.000	A	53	93.201	13%
2	HR	28.800.000				
3	Electricity	2.760.000				
Total		37.560.000				

4.4 EOQ Method

Calculation of spare part inventory using the EOQ method. In calculating inventory using the EOQ method, the optimal number of

purchases, the amount of safety stock, the number of reorder points and the maximum amount of storage will be calculated.

$$EOQ = \frac{\sqrt{2 \times S \times D}}{H} = \frac{\sqrt{2 \times S \times D}}{I \times C}$$

Ordering cost (S) = 29,777

Storage costs (I) = 13 %

Quantity required (D) = 1,856

Purchases per unit (C) = 56,929

For example, the EOQ calculation of the top spare part item with item type (MPX2 10W30 SL 0.8L IDE OIL).

$$EOQ = \frac{\sqrt{2 \times S \times D}}{H} = \frac{\sqrt{2 \times S \times D}}{I \times C}$$

$$EOQ = \frac{\sqrt{2 \times 29,777 \times 1,856}}{13\%}$$

$$EOQ = \sqrt{14,935}$$

$$EOQ = 122,2 \text{ or } 122 \text{ ordering units.}$$

For group A ordering frequency based on item type units is calculated as follows:

$$F = D/Q$$

Where :

D = Amount of Request

Q = Order Quantity

$$F = 1,856/122 = 15$$

So the frequency of placing orders using the EOQ method is 15 orders.

4.5 Safety Stock

Safety stock is inventory carried out to anticipate uncertainty in demand and shortages of spare parts. Calculated using the formula:

$$SS = \frac{LT}{\text{Working Days}} \times D$$

Where :

SS: Safety Stock

LT : 10 days (waiting time)

Working Days: 26 Days (1 month)

D : 155 (demand quantity per month)

For example, calculating safety stock from spare parts with a predetermined lead time of 10 days, working days in one month are 26 days and monthly demand quantity is 155 units. An example of this calculation for the item type (OLI MPX2 10W30 SL 0.8L IDE) is as follows:

$$SS = \frac{10}{26} \times 1,856 \text{ (annual qty) / 12 month}$$

$$SS = \frac{10}{26} \times 155$$

$$SS = 59$$

So the safety stock for the item type unit in

OLI MPX2 10W30 SL 0.8L IDE is 59 units.

4.6 Reorder Point

If the company has a waiting time of 10 days with a safety stock of 59 and daily usage of 5.23 or 5 units, then:

$$ROP = (l \times D) + SS$$

Where:

ROP = Reorder point

D = 5 (Number of Uses Per Day)

l = 10 (Lead time)

SS = 59 (Safety stock)

As an example of calculating the reorder point for spare parts with item type (MPX2 10W30 SL 0.8L IDE OIL) on AHASS Arie Indah Motor as follows:

$$ROP = (l \times D) + SS$$

$$ROP = (10 \times (1,856/355 (1 \text{ Year})) + 59$$

$$ROP = (10 \times 5) + 59$$

$$ROP = 112 \text{ Units}$$

So the ROP (reorder point) for the item type unit of OLI MPX2 10W30 SL 0.8L IDE is 112 units.

4.7 Maximum Inventory

This calculation is carried out in order to find out how much maximum inventory is in the warehouse.

$$Max = EOQ + SS$$

Where:

EOQ = Optimal purchase quantity

SS = Safety stock

For example, the max inventory calculation of spare parts with item type (MPX2 10W30 SL 0.8L IDE OIL) on AHASS Arie Indah Motor is as follows:

$$Max = EOQ + SS$$

$$Max = 122 + 59$$

$$Max = 182 \text{ Units}$$

So the maximum inventory for each item type in OLI MPX2 10W30 SL 0.8L IDE is 182 units.

5. CONCLUSION

a. ABC Analysis:

In the grouping analysis section, three classes were obtained, namely A, B and C.

Class A is a priority because it has a high investment value of 80% and of the 403 total types of goods, 53 types of goods are included in category A. Category B has 92 types and category C has 258 types.

- b. Order Cost :
Each of these costs has been set at an annual value of IDR. 12,000,000 divided by the total types of goods, namely 403, and we get an ordering cost of Rp. 29,777.
- c. Storage Cost:
These costs include depreciation, human resources, and electricity. Calculated with a total annual value of IDR 37,560,000. Obtained a storage value of 93,201 and a percentage of 13% calculated in category A priority goods.
- d. EOQ calculation :
Analysis using the EOQ method obtained a value or unit as a reference point in managing inventory when the company should place an order and when the company should maintain inventory. From the calculation results, the EOQ value is 122 units with a frequency of 15 orders for one type of item (OLI MPX 10W30 SL 0.8L IDE).
- e. Safety Stock :
This calculation is the point at which inventory items must be maintained so as not to experience shortages or empty stocks when the company experiences an increase in demand. In determining safety stock, the company sets a waiting time and determines the quantity of goods needed. The calculation results obtained a safety stock of 59 units type of item (OLI MPX 10W30 SL 0.8L IDE).
- f. Reorder Point (ROP) :
The calculations that have been carried out have resulted in 122 units with a waiting time of 10 days and an average demand of 5 units per day. This calculation is an example of one type of spare part, namely (MPX 10W30 SL 0.8L IDE OIL).
- g. Maximum Inventory :
The maximum inventory can be found by adding the results of the EOQ calculation with the safety stock, then the maximum value will be obtained. From the results of the addition of the above Maximum Inventory, it is obtained that there are 182.

6. REFERENCES

- Abdullah, P. M. (2015). Living in the world that is fit for habitation: CCI's ecumenical and religious relationships. *In Aswaja Pressindo*
- Al-Fikri, Z., & Ghofari, A. K. Al. (2023). Analisis Pengendalian Persediaan *Sparepart* Menggunakan Pendekatan Metode ABC dan EOQ (Studi Kasus PT. Solusi Bangun Indonesia). *Doctoral Dissertation, Universitas Muhammadiyah Surakarta.*, 5–24. <http://repo.iainlungagung.ac.id/5510/5>
- Guslan, D., & Saputra, I. (2020). Analisis Pengendalian *Inventori* Dengan Klasifikasi ABC dan EOQ Pada PT Nissan Motor Distributor Indonesia. *Jurnal Logistik Bisnis*, 10(1), 73. <https://doi.org/10.46369/logistik.v10i1.700>.
- Hakim, Z., Sakuroh, L., & Awaludin, S. (2019). Sistem Informasi Persediaan Barang Berbasis Web Pada CV Telaga Berkat. *Jurnal Sisfotek Global*, 9(1). <https://doi.org/10.38101/sisfotek.v9i1.214>.
- Heizer, J. dan Render, B.2015. *Operasional Management*. Edisi Sebelas. Salemba Empat, Jakarta.
- Hidayat, K., Efendi, J., & Faridz, R. (2020). Analisis Pengendalian Persediaan Bahan Baku Kerupuk Mentah Potato Dan Kentang Keriting Menggunakan Metode *Economic Order Quantity (EOQ)*. *Performa: Media Ilmiah Teknik Industri*, 18(2), 125–134. <https://doi.org/10.20961/performa.18.2.35418>.
- Lutfiana, L., & Puspitosari, I. (2020). Analisis Manajemen Persediaan Pada Usaha Mikro, Kecil, dan Menengah (UMKM) Jazid Bastomi Batik Di Purworejo *Inventory Management Analysis Of Small and Medium Enterprises. Jurnal JESKape*, 4(1), 55–66.
- Mayasari, D. (2021). Analisis Pengendalian Persediaan Bahan Baku Menggunakan Metode EOQ (*Economic Order Quantity*) pada PT. Suryamas Lestari Prima. *Bis-a*, 10(02), 44-50.
- Meirizha, S. N. (2022). Penerapan Metode *Economic Order Quantity (EOQ)* pada

- Persediaan *Spare Part* Jenis Oli PT. Agung Automall Sutomo. *Jurnal Surya Teknik*, 9(2), 492–497. <https://doi.org/10.37859/jst.v9i2.4392>.
- Ramadhan, M. F. (2018). Pengendalian Persediaan Bahan Baku Daun Teh Broken Pecco 1 Menggunakan Metode *EOQ (Economic Order Quantity)* pada Perusahaan Agroindustri PT XYZ. 1–118.
- Safitri, L. (2021). Analisis Pengendalian Persediaan Bahan Baku Melta Vanana Dengan Menggunakan Metode *Economic Order Quantity (EOQ)* Pada CV. Vanana Jaya Sinergi. *Angewandte Chemie International Edition*, 6(11), 951–952., 5–24.
- Sofiana, A., & Ahmad Tasdiqul Haq, D. (2020). Pengendalian Persediaan Insert Tools dengan Metode *Economic Order Quantity* dan Klasifikasi *ABC*. *Jurnal INTECH Teknik Industri Universitas Serang Raya*, 6(1), 39–47. <https://doi.org/10.30656/intech.v6i1.2174>.
- Sudaryana, Bambang D. E. A., Ak, M., Agusiady, H. R., & SE, M. (2022). *Metodologi penelitian kuantitatif*. Deepublish.
- Sugiyono. 2019. Metode Penelitian Kuantitatif, Kualitatif dan R & D. Bandung: Alfabeta
- Supriyadi, E., & Nurdewanti, R. (2022). Pengendalian Persediaan Bahan Baku dengan Metode *Always Better Control (ABC)* dan *Economic Order Quantity (EOQ)* di CV. XYZ. *Briliant: Jurnal Riset dan Konseptual*, 7(1), 211. <https://doi.org/10.28926/briliant.v7i1.888>.